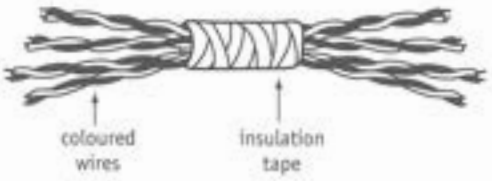
Solving problems electrically 9a

1 A bundle of several coloured wires, all insulated from each other, is taped together with black insulation tape. Unfortunately one of the wires is broken.



Design an experiment to find out which coloured wire is broken.

Title: _____

Aim: _____

Equipment: _____

_____

Method: _____

_____

_____

_____

Diagram:

Conclusion: _____

_____

A student has built a quiz box, which is a question and answer game. This is a small box, which lists six questions on the left side and six answers on the right side (see box below). The end of a wire is connected to a contact behind each of the questions. The other end of the wire goes to a bell and a battery, both connected in series. Another wire comes off the battery and is connected to a contact in front of the corresponding answer. If a player selects the correct answer, the bell rings. If the selected answer is incorrect, nothing at all happens.

2 Match the questions on the left side of the box to the correct answers on the right side.

Questions		Answers
a Potential difference is measured in ... -		- ... amps
b Resistance is calculated by ... -		- ... amps
c Ohm's Law is about ... -		- ... dividing volts by amps
d The unit of current is ... -		- ... resistance
e A fuse melts when there are too many ... -		- ... volts
f House mains provide 240 ... -		- ... volts

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3 Show what the game would look like in action by sketching the bell, battery, wires and box.

4 The student who built the quiz box had connected the contacts of the questions with contacts of the answers with wires inside the quiz box. Add those wires to your sketch of the box in question 2.

5 Design a similar question and answer box. Write your questions and answers in the table below. Then swap your design with someone in the class and figure out the wiring inside each other's quiz box by connecting the questions to the correct answers.

Questions		Answers

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